

The Exact Counting Complexity of $\#EO(\{f_{56}\})$

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Abstract

We give a self-contained proof that the exact counting problem $\#EO(\{f_{56}\})$ is $\#P$ -complete under polynomial-time Turing reductions. The argument has three steps. First, the signature f_{56} admits an exact reformulation as a fixed binary counting CSP on a five-element domain. Second, after adjoining equality and deleting the unique empty row relation, we exhibit an explicit pp-definable ternary relation whose balance matrix is

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

so the resulting language is not strongly balanced in the sense of Dyer and Richerby. Their dichotomy theorem therefore yields $\#P$ -completeness for the auxiliary language. Third, we show that the auxiliary problem polynomial-time Turing reduces back to the original raw language by contracting equality constraints and replacing the resulting unary constraints by fixed raw gadgets with a known constant multiplicative factor. Combining these ingredients gives the $\#P$ -completeness of $\#EO(\{f_{56}\})$.

1 Introduction

The signature f_{56} is the remaining exceptional case singled out in the current EO-dichotomy program of Meng, Wang, and Xia [2, 3]. The main result of this paper is that the exact counting problem for this signature is in fact $\#P$ -complete.

Theorem 1.1. *The exact counting problem $\#EO(\{f_{56}\})$ is $\#P$ -complete under polynomial-time Turing reductions.*

Throughout the paper, all counting-completeness statements are understood with respect to polynomial-time Turing reductions. The nontrivial part is the hardness proof.

2 The Signature f_{56} and Its Raw Binary Language

2.1 Definition of the signature

Let

$$D = [5] = \{1, 2, 3, 4, 5\}.$$

The support matrix of f_{56} is the 5×56 matrix whose columns are indexed by the multiset

$$\{\emptyset\} \cup \bigsqcup_{|S|=2} S^{\times 4} \cup \bigsqcup_{|S|=4} S^{\times 3},$$

and whose entry in row $i \in D$ and column type S is $\mathbf{1}_{i \in S}$. Equivalently, this is the support matrix denoted

$$[H_0 \ H_2^{\rightarrow 4} \ H_4^{\rightarrow 3}]$$

in the literature [3]. The signature f_{56} is the 0-1-valued EO signature whose support consists exactly of the five row strings of this matrix.

2.2 Exact binary-CSP reformulation

For even subsets $S, T \subseteq D$ with $|S|, |T| \in \{0, 2, 4\}$, define the binary relation

$$R_{S,T}(x, y) = 1 \iff \mathbf{1}_{x \in S} \oplus \mathbf{1}_{y \in T} = 1.$$

Let

$$\Gamma_{\text{raw}} := \{R_{S,T} : S, T \subseteq D, |S|, |T| \in \{0, 2, 4\}\}.$$

Definition 2.1. An instance of $\#\text{CSP}(\Gamma)$ for a finite relation language Γ consists of a finite set of variables together with constraints of the form

$$H(x_{i_1}, \dots, x_{i_r}), \quad H \in \Gamma.$$

Its value is the number of satisfying assignments over the underlying finite domain.

Lemma 2.2 (Exact reformulation). *Every instance of $\#\text{EO}(\{f_{56}\})$ is exactly an instance of $\#\text{CSP}(\Gamma_{\text{raw}})$, with one variable $x_v \in D$ for each vertex v , and one binary constraint $R_{S,T}$ for each edge joining a half-edge of type S at one endpoint to a half-edge of type T at the other.*

Equivalently,

$$\#\text{EO}(\{f_{56}\}) = \#\text{CSP}(\Gamma_{\text{raw}}).$$

Proof. At each vertex, choosing a locally valid configuration of f_{56} is the same thing as choosing one of the five rows, i.e. one label $i \in D$.

Consider a half-edge position of type S . In row i , the local bit on that half-edge is exactly $\mathbf{1}_{i \in S}$. If an edge joins a half-edge of type S at vertex u to a half-edge of type T at vertex v , then an Eulerian orientation puts opposite bits on the two ends of that edge: one endpoint gets bit 0 and the other gets bit 1. Hence a pair of row choices (x_u, x_v) is feasible on that edge if and only if

$$\mathbf{1}_{x_u \in S} \oplus \mathbf{1}_{x_v \in T} = 1,$$

which is exactly the relation $R_{S,T}$.

Since f_{56} is 0-1-valued, the partition function counts the row-labelings $(x_v)_v \in D^V$ satisfying all these edge relations. This is precisely $\#\text{CSP}(\Gamma_{\text{raw}})$. $\square$

3 An Auxiliary Language Compatible with Dyer–Richerby

Dyer and Richerby [1] define a constraint language to be finite, to contain equality, and to have no empty basic relations. The raw language Γ_{raw} misses equality and contains one empty relation, namely $R_{\emptyset, \emptyset}$. We therefore pass to a closely related auxiliary language.

Definition 3.1. Define

$$\Gamma_{\text{dr}} := (\Gamma_{\text{raw}} \setminus \{R_{\emptyset, \emptyset}\}) \cup \{=\}.$$

Lemma 3.2. Γ_{dr} is a finite relation language on D that contains equality and whose basic relations are all nonempty.

Proof. By construction, Γ_{dr} is finite and contains equality. It remains to identify the empty raw relations.

For any $S, T \subseteq D$,

$$R_{S,T}(x, y) = 1 \iff \mathbf{1}_{x \in S} \oplus \mathbf{1}_{y \in T} = 1.$$

This predicate is identically false exactly when both membership indicators are always 0, which happens if and only if $S = T = \emptyset$. Hence the unique empty relation in Γ_{raw} is $R_{\emptyset, \emptyset}$, and removing it gives a nonempty language. $\square$

3.1 Unary relations from raw gadgets

Lemma 3.3 (Constant implementations of unary relations). *For every unary relation $U \subseteq D$, there is a primitive-positive formula $\phi_U(x, \mathbf{z})$ over Γ_{raw} such that:*

- (1) *if $x \in U$, then exactly $5^{|D \setminus U|}$ assignments to $\mathbf{z}$ satisfy $\phi_U(x, \mathbf{z})$;*
- (2) *if $x \notin U$, then no assignment to $\mathbf{z}$ satisfies $\phi_U(x, \mathbf{z})$.*

In particular, every unary relation on D is pp-definable over both Γ_{raw} and Γ_{dr} .

Proof. Fix $U \subseteq D$. For each $j \in D \setminus U$, let $\bar{j} := D \setminus \{j\}$. Since $|\emptyset| = 0$ and $|\bar{j}| = 4$, the relation $R_{\emptyset, \bar{j}}$ belongs to Γ_{raw} .

For every $x, z \in D$,

$$R_{\emptyset, \bar{j}}(z, x) = 1 \iff \mathbf{1}_{z \in \emptyset} \oplus \mathbf{1}_{x \in \bar{j}} = 1 \iff x \in \bar{j} \iff x \neq j.$$

So $R_{\emptyset, \bar{j}}(z, x)$ is independent of z : if $x \neq j$, it is true for all 5 choices of z , and if $x = j$, it is false for all choices of z .

Now set

$$\phi_U(x, \mathbf{z}) := \bigwedge_{j \in D \setminus U} R_{\emptyset, \bar{j}}(z_j, x).$$

If $x \in U$, then every conjunct is satisfied and each witness variable z_j can be chosen arbitrarily, yielding exactly $5^{|D \setminus U|}$ satisfying witness assignments. If $x \notin U$, then $x = j$ for some $j \in D \setminus U$, so the corresponding conjunct is false for every z_j . This proves both claims. $\square$

4 The Dyer–Richerby Criterion

Definition 4.1. Let A_1, A_2 be finite sets, and let $M : A_1 \times A_2 \rightarrow \mathbb{Z}_{\geq 0}$ be a matrix. We say that M is a *rank-one block matrix* if its support

$$\{(x, y) \in A_1 \times A_2 : M(x, y) > 0\}$$

is a disjoint union of rectangles and on each connected support block the corresponding submatrix has rank at most 1.

Definition 4.2. Let $H \subseteq A_1 \times A_2 \times A_3$ be a ternary relation. Its *balance matrix* is the matrix

$$M_H(x, y) = |\{z \in A_3 : (x, y, z) \in H\}|.$$

We say that H is *balanced* if M_H is a rank-one block matrix. A finite constraint language Γ is *strongly balanced* if every pp-definable ternary relation over Γ is balanced.

Theorem 4.3 (Dyer–Richerby). *If a finite constraint language Γ is strongly balanced, then $\#\text{CSP}(\Gamma)$ is in FP. Otherwise $\#\text{CSP}(\Gamma)$ is $\#P$ -complete.*

This is exactly Theorem 28 of Dyer and Richerby [1].

5 An Explicit Unbalanced Ternary Relation

Lemma 5.1. *There exists a pp-definable ternary relation H over Γ_{dr} that is not balanced.*

More precisely, let

$$U_{12} := \{1, 2\} \subseteq D,$$

and define

$$H(x, y, z) \iff U_{12}(x) \wedge U_{12}(y) \wedge R_{\{1,3\}, \{1,2\}}(x, z) \wedge R_{\{1,3\}, \{1,3\}}(y, z).$$

Then the balance matrix of H on $A_1 = A_2 = \{1, 2\}$ and $A_3 = D$ is

$$M_H = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

In particular, H is not balanced.

Proof. By Lemma 3.3, the unary relation U_{12} is pp-definable over Γ_{raw} , hence also over Γ_{dr} . Thus the displayed formula indeed defines a pp-definable ternary relation.

We now compute the allowed z -values.

For $x = 1$, since $1 \in \{1, 3\}$,

$$R_{\{1,3\},\{1,2\}}(1, z) = 1 \iff 1 \oplus \mathbf{1}_{z \in \{1,2\}} = 1 \iff z \in \{3, 4, 5\}.$$

For $x = 2$, since $2 \notin \{1, 3\}$,

$$R_{\{1,3\},\{1,2\}}(2, z) = 1 \iff \mathbf{1}_{z \in \{1,2\}} = 1 \iff z \in \{1, 2\}.$$

For $y = 1$, since $1 \in \{1, 3\}$,

$$R_{\{1,3\},\{1,3\}}(1, z) = 1 \iff 1 \oplus \mathbf{1}_{z \in \{1,3\}} = 1 \iff z \in \{2, 4, 5\}.$$

For $y = 2$, since $2 \notin \{1, 3\}$,

$$R_{\{1,3\},\{1,3\}}(2, z) = 1 \iff \mathbf{1}_{z \in \{1,3\}} = 1 \iff z \in \{1, 3\}.$$

Therefore

$$\begin{aligned} \{z : H(1, 1, z)\} &= \{3, 4, 5\} \cap \{2, 4, 5\} = \{4, 5\}, \\ \{z : H(1, 2, z)\} &= \{3, 4, 5\} \cap \{1, 3\} = \{3\}, \\ \{z : H(2, 1, z)\} &= \{1, 2\} \cap \{2, 4, 5\} = \{2\}, \\ \{z : H(2, 2, z)\} &= \{1, 2\} \cap \{1, 3\} = \{1\}. \end{aligned}$$

So indeed

$$M_H = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its support is the full 2×2 rectangle, and the determinant is $2 \cdot 1 - 1 \cdot 1 = 1$, so this support block has rank greater than 1. Hence H is not balanced. $\square$

Corollary 5.2. $\#\text{CSP}(\Gamma_{\text{dr}})$ is $\#P$ -complete.

Proof. By Lemma 3.2, the language Γ_{dr} satisfies the conventions required by Theorem 4.3. By Lemma 5.1, Γ_{dr} has a pp-definable ternary relation that is not balanced. Therefore Γ_{dr} is not strongly balanced, and Theorem 4.3 implies that $\#\text{CSP}(\Gamma_{\text{dr}})$ is $\#P$ -complete. $\square$

6 Reducing Back to the Raw Language

Lemma 6.1. $\#\text{CSP}(\Gamma_{\text{dr}})$ polynomial-time Turing reduces to $\#\text{CSP}(\Gamma_{\text{raw}})$.

Proof. Let I be an instance of $\#\text{CSP}(\Gamma_{\text{dr}})$.

First contract each connected component of the equality constraints in I to a single variable. This preserves the set of satisfying assignments. After contraction, every remaining equality constraint is redundant and can be deleted.

Every surviving non-equality constraint then has one of two forms:

- (1) a binary raw constraint $R_{S,T}(x, y)$ with $x \neq y$;

(2) a unary self-loop constraint $U(x)$, where

$$U = \{a \in D : R_{S,T}(a, a) = 1\}$$

for some $R_{S,T} \in \Gamma_{\text{raw}} \setminus \{R_{\emptyset, \emptyset}\}$.

Thus we obtain an equivalent instance I' consisting of binary raw constraints and unary constraints on D .

Now replace each unary constraint $U(x)$ in I' by the raw gadget $\phi_U(x, \mathbf{z})$ from Lemma 3.3. Suppose the unary constraints are $U_1(x_1), \dots, U_m(x_m)$, and let

$$c := \prod_{i=1}^m 5^{|D \setminus U_i|}.$$

By Lemma 3.3, every satisfying assignment of I' extends to exactly c satisfying assignments of the resulting raw instance J , and every satisfying assignment of J restricts to a satisfying assignment of I' . Hence

$$\#J = c \cdot \#I'.$$

Therefore

$$\#I = \#I' = \frac{\#J}{c}.$$

The transformation $I \mapsto J$ and the constant c are computable in polynomial time because the domain and language are fixed. This yields the claimed polynomial-time Turing reduction. $\square$

7 The Main Hardness Theorem

Theorem 7.1. $\#\text{CSP}(\Gamma_{\text{raw}})$ is $\#P$ -complete.

Proof. Corollary 5.2 shows that $\#\text{CSP}(\Gamma_{\text{dr}})$ is $\#P$ -complete. By Lemma 6.1, this problem polynomial-time Turing reduces to $\#\text{CSP}(\Gamma_{\text{raw}})$. Therefore $\#\text{CSP}(\Gamma_{\text{raw}})$ is $\#P$ -hard.

On the other hand, $\#\text{CSP}(\Gamma_{\text{raw}})$ plainly belongs to $\#P$: it counts satisfying assignments of a fixed finite-domain relation CSP instance. Hence it is $\#P$ -complete. $\square$

Proof of Theorem 1.1. By Lemma 2.2,

$$\#\text{EO}(\{f_{56}\}) = \#\text{CSP}(\Gamma_{\text{raw}}).$$

Therefore Theorem 7.1 implies that $\#\text{EO}(\{f_{56}\})$ is $\#P$ -complete. $\square$

Remark 7.2. The only place where we appeal to external counting-CSP theory is Theorem 4.3. All other ingredients in the proof are explicit and special to the concrete raw language induced by f_{56} .

References

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- [2] Boning Meng, Juqiu Wang, and Mingji Xia, *P-time Algorithms for Typical #EO Problems*, arXiv:2410.11557, 2024.
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